

Advanced Optimization — Lecture Plan

Fall 2026, Sogang University, Dept. of Electronic Engineering Prof. Hongseok Kim

Text: Nocedal & Wright, *Numerical Optimization*, 2nd ed. Tue/Thu. Revised Sep 22, 2026.

Scope. The course covers *unconstrained* optimization in depth (Ch. 3–7) and the *theory* of constrained optimization (Ch. 12). It closes with a two-lecture reading guide to primal-dual interior-point methods (Ch. 19) — the algorithm inside IPOPT and KNITRO — and a final project in which you run IPOPT on a small AC OPF and explain every line of its log using Ch. 3, 12 and 19. Constrained *algorithms* (Ch. 15–18) are outside the course; the last section says when you would need them.

Wk	Date	Topic	Reading / HW
1–4	Sep 1–22	Ch. 1–2; Ch. 3 line search (Wolfe, Zoutendijk, Newton/quasi-Newton rates, Hessian modification); Ch. 4 framework, Algorithm 4.1, Theorem 4.1 — <i>done</i>	HW3 out Sep 22
4	Sep 24	<i>No class (Chuseok)</i>	
5	Sep 29	Ch. 4: Cauchy point, fraction of Cauchy decrease, dogleg, 2D subspace minimization; Steihaug-CG named only	§4.1
	Oct 1	Ch. 4: global convergence (Thm. 4.5–4.6), local convergence, scaling; §4.3 skipped	§4.2, 4.4–4.5; HW4 out
6	Oct 6	Ch. 5: linear CG I — conjugate directions, expanding subspace (Thm. 5.2), Algorithm 5.1 → 5.2	§5.1
	Oct 8	Ch. 5: linear CG II — rate of convergence, eigenvalue clustering, preconditioning; nonlinear CG (FR, PR ⁺ , Thm. 5.7 and why $c_2 < \frac{1}{2}$); Steihaug-CG properly	§5.1–5.2, §4.1; HW5 out
7	Oct 13	Ch. 6: BFGS — secant equation, why $s^T y > 0$ needs Wolfe, implementation	§6.1
	Oct 15	Ch. 6: SR1, Broyden class, convergence results (stated), Dennis–Moré revisited	§6.2–6.4; HW6 out
8	Oct 20	Midterm exam (Ch. 3–6)	
	Oct 22	Buffer / catch-up (midterm week)	
9	Oct 27	Ch. 7: inexact Newton, Newton-CG, trust-region Newton-CG	§7.1
	Oct 29	Ch. 7: L-BFGS; what is skipped in §7.3–7.4 and why	§7.2; HW7 out
10	Nov 3	Ch. 12: examples, tangent cone, LICQ, first-order necessary conditions (statement)	§12.1–12.2
	Nov 5	Ch. 12: proof of KKT via Farkas’ lemma; constraint qualifications	§12.3, 12.6
11	Nov 10	Ch. 12: second-order conditions, critical cone	§12.5
	Nov 12	Ch. 12: degeneracy, duality, multipliers as sensitivities	§12.7–12.9; HW12 out
12	Nov 17	Ch. 12 in practice: KKT of a small AC OPF; multipliers as prices	notes
	Nov 19	Buffer / Ch. 12 spillover	
13	Nov 24	Ch. 19 reading guide I: barrier problem, primal-dual Newton system, central path, inertia (§3.4 again)	§19.1–19.3
	Nov 26	Ch. 19 reading guide II: merit function / filter line search; an IPOPT log, line by line	§19.4–19.6; Wächter–Biegler (2006)
14	Dec 1	Final project workshop (IPOPT via JuMP / Pyomo / CasADi)	
	Dec 3	Buffer / review	
15	Dec 8, 10	Final project presentations	
16	Dec 15, 17	Final exam week — no lecture; project report due	

Assumed: 16-week semester from Sep 1, midterm week 8 (Oct 20–22), final week 16 (Dec 15–17). Adjust if the official calendar differs.

Homework thread. One problem set per chapter, one week each. Code is expected to be written with AI

assistance; grading is on the *observations*, not the code. The line search from HW3 is reused in HW5–HW7.

- **HW3** — backtracking (Alg. 3.1) and strong-Wolfe (Alg. 3.5–3.6) line searches; steepest descent and Newton on Rosenbrock; Alg. 3.3 on an indefinite Hessian; Zoutendijk annotated; Ex. 3.5.
- **HW4** — Algorithm 4.1 with dogleg on Rosenbrock; $\log \Delta_k, \rho_k$; compare iteration counts with HW3.
- **HW5** — linear CG on an SPD matrix with clustered vs. spread eigenvalues ($n = 100$): iteration count follows the clusters; FR and PR^+ with the HW3 line search, $c_2 = 0.1$.
- **HW6** — BFGS with the HW3 line search on Rosenbrock; $\log \|B_k - \nabla^2 f_k\|$ against $\|(B_k - \nabla^2 f_k)p_k\|/\|p_k\|$ (Dennis–Moré in action).
- **HW7** — L-BFGS with $m \in \{3, 5, 20\}$ on an $n = 10^4$ problem; compare with Newton-CG.
- **HW12** — KKT conditions by hand for a 2- or 3-bus DC OPF; identify what each multiplier means.

Final project. Solve a small AC OPF (pglib-opf **case5** or **case14**) with IPOPT. Annotate the solver log: barrier parameter μ and its decrease (§19.3), primal/dual infeasibility and complementarity (Ch. 12), **inertia correction** (§3.4), step lengths α and the merit/filter decision (§19.4). Report: 4 pages, plus a 10-minute presentation.

Not covered, and when you would need it.

- Ch. 8 (derivatives, automatic differentiation) — this is what **autograd** does; read §8.2 once.
- Ch. 9 (derivative-free), Ch. 11 (nonlinear equations) — niche for this course.
- Ch. 10 (nonlinear least squares: Gauss–Newton, Levenberg–Marquardt) — read if you fit models; it is Ch. 4 applied.
- Ch. 13 (simplex), Ch. 14 (LP interior point) — Ch. 14’s idea is all that Ch. 19 needs, and it is given in the reading guide.
- Ch. 15–18 (merit functions, QP, penalty/augmented Lagrangian, SQP) — read Ch. 17 if you use ADMM-type methods; Ch. 18 is KNITRO’s other mode.